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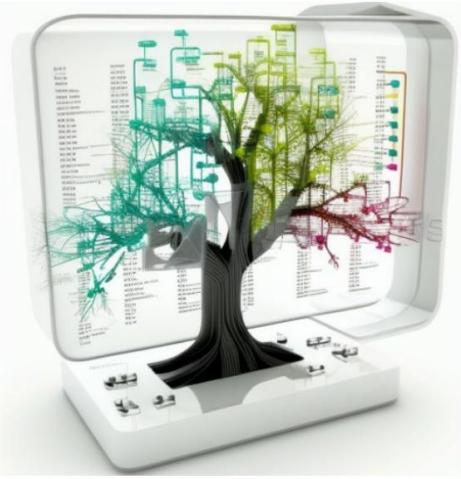
[Chap 8] Decision Tree



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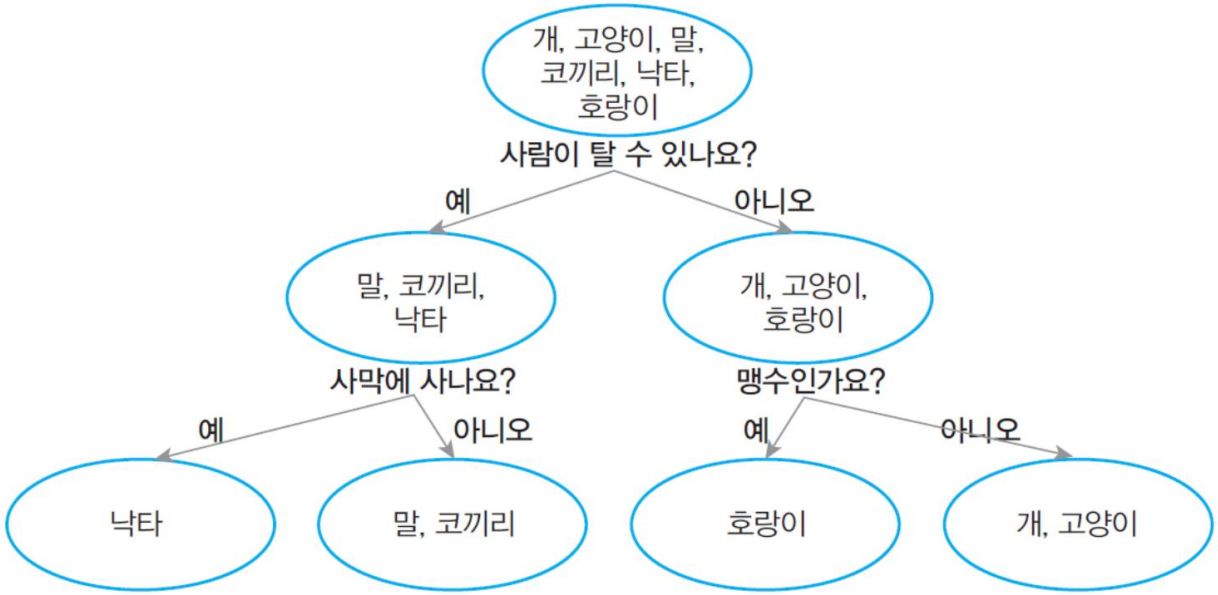
Twenty Questions

- Consists of simple binary classification
  - Branch to yes or no
- The decision tree model is the same concept.



[그림 8-1] 스무고개 놀이

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Decision Tree

- Represent decision rules in a tree structure to classify the entire data into several subgroups

  Analysis methods that perform (classification) or prediction

- Key Term –

  Node or joint • The numbers

  1 through 9 in the figure – Root node • A node

  that only starts and does

  not come from anywhere else (No. 1 in the figure)

  – Terminal node or leaf node • Nodes that do not branch further

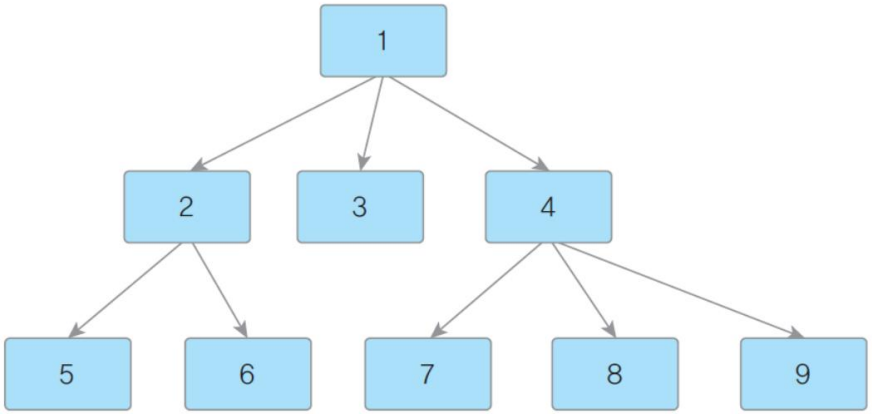
  (5, 6, 7, 8, 9 in the figure) – Parent node – Child

  node

- A structure where the number of

  corresponding data decreases

  as the branching continues.



[그림 8-2] 의사결정나무 예시

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Decision Tree Practice

- Kaggle's data on predicting customer churn in bank credit card usage
  - bank_customer.csv

〈표 8-1〉 은행 신용카드 고객 데이터 열 설명

열 이름	설명
CLIENTNUM	고객번호
Attrition_Flag	이탈고객여부. 0=유지, 1=이탈
Customer_Age	고객 나이
Gender	고객 성별. 0=남성, 1=여성
Dependent_count	부양가족 수
Income_Category	연소득 수준. 0=자료없음, 1=$40K 미만, 2=$40K - 60K, 3=$60K - $80K, 4=$80K-$120K, 5=$120K 초과
Card_Category	카드 등급. 0=Blue, 1=Silver, 2=Gold, 3=Platinum
Months_on_book	은행과의 총 거래기간
Total_Relationship_Count	은행 상품 보유 갯수
Contacts_Count_12_mon	최근 12개월간 은행과의 접촉 수
Credit_limit	신용 한도
Total_Revolving_Bal	리볼빙 잔액
Avg_Open_To_Buy	신용 잔여액
Total_Trans_Amt	평균 거래금액
Total_Amt_Chng_Q4_Q1	거래금액 변화량 (Q4 대비 Q1)

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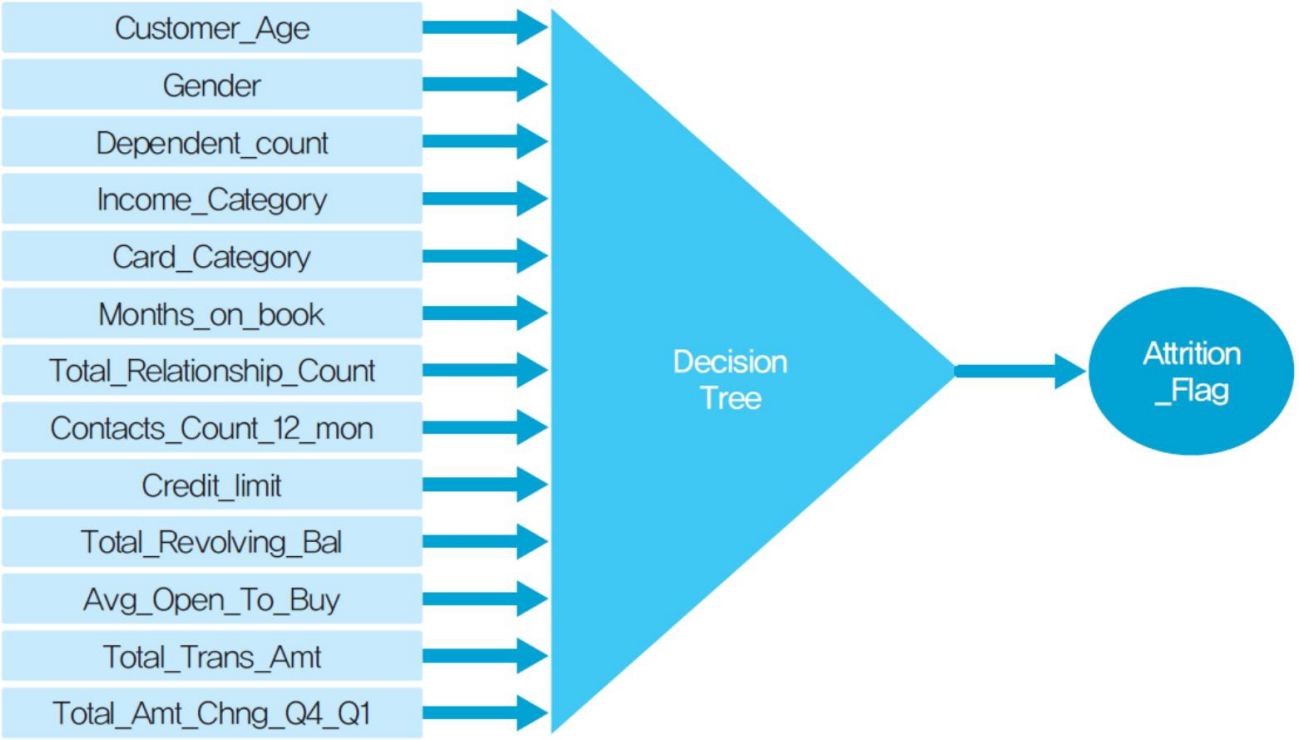
Features and Targets



[그림 8-3] 은행 신용카드 고객 이탈 예측을 위한 모형

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Read data

- Read bank_customer.csv to construct the training data structure

```
1  import pandas as pd
2  df = pd.read_csv('bank_customer.csv')
3  feature = df[['Customer_Age', 'Gender', 'Dependent_count', 'Income_Category',
4    'Card_Category', 'Months_on_book', 'Total_Relationship_Count',
5    'Contacts_Count_12_mon', 'Credit_Limit', 'Total_Revolving_Bal',
6    'Avg_Open_To_Buy', 'Total_Trans_Amt', 'Total_Amt_Chng_Q4_Q1']]
7  target = df['Attrition_Flag']
```

– (Rows 3–6) Select columns from the DataFrame to be used as feature values and store them in the feature variable –

(Row 7) Store the correct answer column 'Attrition_Flag' in the target variable

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Training data and test data

- Divide the data into training data and test data

```
1  from sklearn.model_selection import train_test_split
2  x_train, x_test, y_train, y_test = train_test_split(feature, target, test_size=0.2,
3  random_state=2222)
```

– (Line 1) Imports the train_test_split function from the sklearn.model_selection

package – (Lines 2–3) Train_test_split contains the feature dataset and the ground truth dataset. Pass the target and set test_size=0.2 to use 20% of the total dataset as test data.

- Consists of 8,101 training data points and 2,026 test data points out of a total of 10,127 data points
- random_state sets the seed value for the random number values to be used when randomly dividing the data. If omitted Training and test data are extracted differently each time it is run.

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Decision Tree Model Creation and Prediction

- Create a decision tree model, train it using a training dataset, and Perform predictions using the test dataset

```
1  from sklearn.tree import DecisionTreeClassifier
2  dtc = DecisionTreeClassifier(random_state=2222)
3  dtc = dtc.fit(x_train,y_train)
4  y_pred = dtc.predict(x_test)
```

– (Line 2) Create a DecisionTreeClassifier model and assign it to a variable named dtc

  - Initialize decision tree model
  - To compare results, the random number seed is fixed at random_state=2222.

– (Line 3) Feature values of the training data in the initial decision tree model stored in dtc
Pass x_train and the ground truth y_train and perform machine learning.

– (Line 4) Insert x_test, the feature value of the test data, into dtc, where the trained decision tree is stored.
Hurry up and predict

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Verify model prediction performance

- Compare the y_pred predicted by the decision tree model with the actual value y_test

  Check the model's performance

```
from sklearn.metrics import confusion_matrix
from sklearn.metrics import classification_report
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

```
[[1621   62]
 [  79  264]]
              precision    recall  f1-score   support

           0       0.95      0.96      0.96      1683
           1       0.81      0.77      0.79       343

    accuracy                           0.93      2026
   macro avg       0.88      0.87      0.87      2026
weighted avg       0.93      0.93      0.93      2026
```

– Demonstrated decent performance with an accuracy of 0.93 and an F1-

score of 0.87 – With precision of 0.81, recall of 0.77, and an F1-score of 0.79 for churned customers (correct value 1),

Customer churn can be predicted to a significant degree.

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Decision tree visualization

- Directly visualize and verify the generated decision tree model.

```
1  from sklearn.tree import plot_tree
2  import matplotlib.pyplot as plt
3
4  plt.figure(figsize=(24,10))
5  plot_tree(dtc, feature_names=['Customer_Age', 'Gender',
6    'Dependent_count', 'Income_Category',
7    'Card_Category', 'Months_on_book', 'Total_Relationship_Count',
8    'Contacts_Count_12_mon', 'Credit_Limit', 'Total_Revolving_Bal',
9    'Avg_Open_To_Buy', 'Total_Trans_Amt', 'Total_Amt_Chng_Q4_Q1'],
10   class_names=['Attrited','Existing'], max_depth=4, rounded=True, fontsize=6)
11 plt.show()
```

– (Row 4) Set the total visualization size to (24,10) –

(Rows 5–10) Visualize the decision tree model using plot_tree() • Pass the

decision tree model dtc as an argument, and the feature names to be displayed in feature_names Set as a list

• In class_names, set the titles corresponding to prediction results 0 and 1 to 'Existing' and 'Attrited'. •

max_depth determines the depth of the decision tree to be displayed on

the screen. • Set the node shape to slightly rounded ends (rounded=True) and the font size to 6 points.

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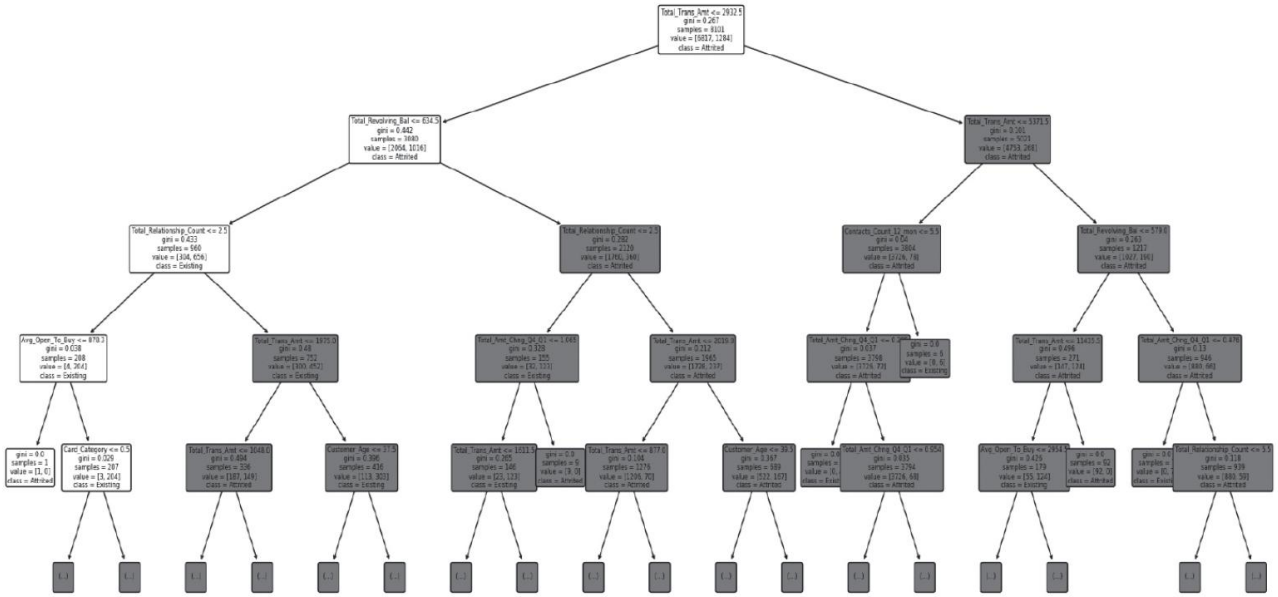
Visualization results

- Each node has a branching criterion written on it – At the root node, based on the transaction amount, it branches to the left if it is less than 2931.5, and to the right otherwise.
  - If you go to the left, it branches again based on whether the revolving balance is less than 634.5 – Branching based on conditions and following the branches to ultimately determine customer churn

- Decision trees have the advantage of being explainable – branching criteria are understandable to humans – widely used in practice



[그림 8–4] 의사결정나무 시각화